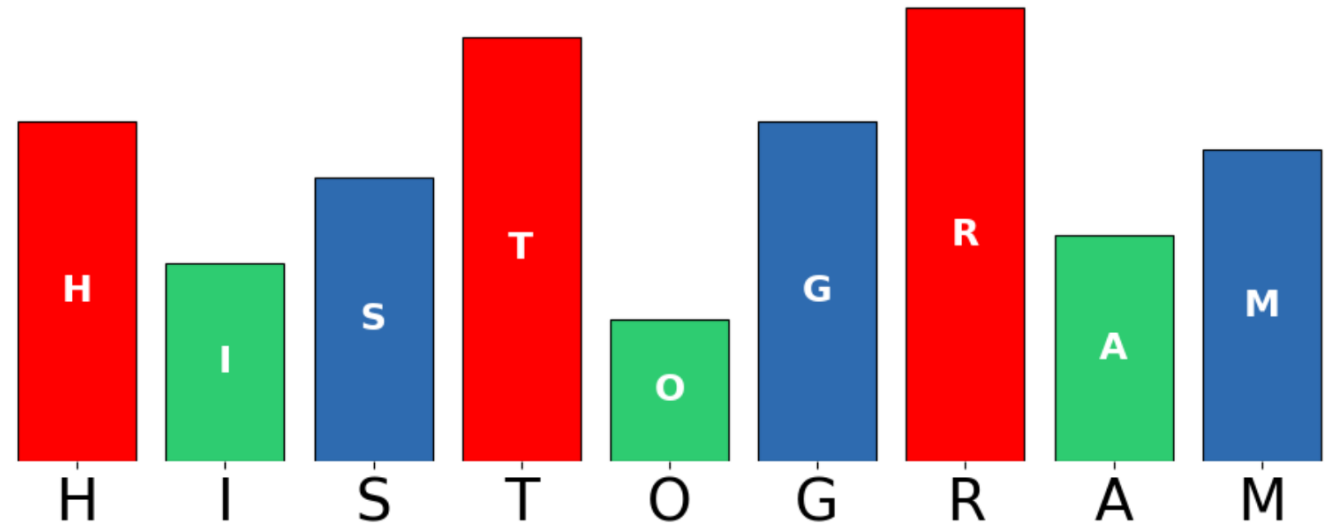




Bagian I

Histogram I

- Thresholding
 - Membuat citra biner → objek mudah dianalisis.
- Equalization
 - Menyebarkan nilai intensitas → histogram lebih merata → meningkatkan kontras.



Histogram Thresholding

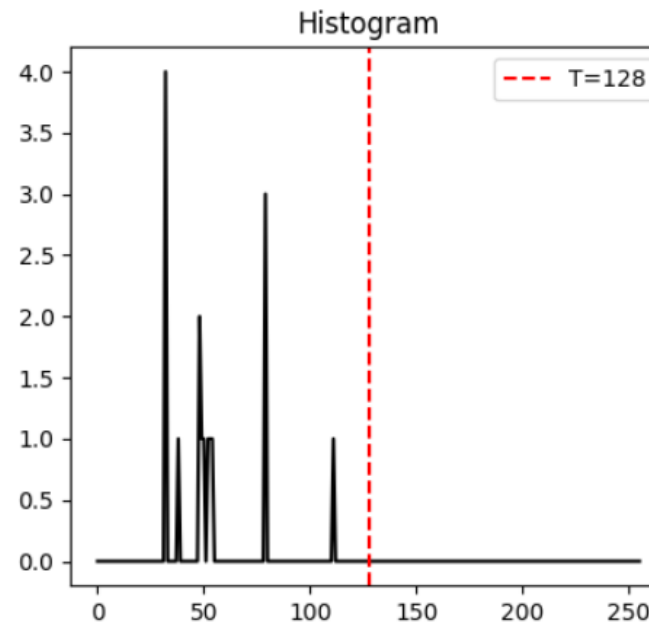
Thresholding (Binary Image)

Dari materi "Operasi Dasar Citra" di perkuliahan sebelumnya, Threshold = 128


$$\begin{bmatrix} 32 & 48 & 32 & 54 \\ 32 & 111 & 79 & 32 \\ 38 & 79 & 79 & 48 \\ 50 & 53 & 52 & 49 \end{bmatrix}$$


Nilai	Freq
32	4
38	1
48	2
50	1
52	1
53	1
54	1
79	3
111	1

Threshold = 128


$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hasil Thresholding



Thresholding (Binary Image) – Nilai Tengah

Cukup gunakan greyscale citra tidak perlu RGB.

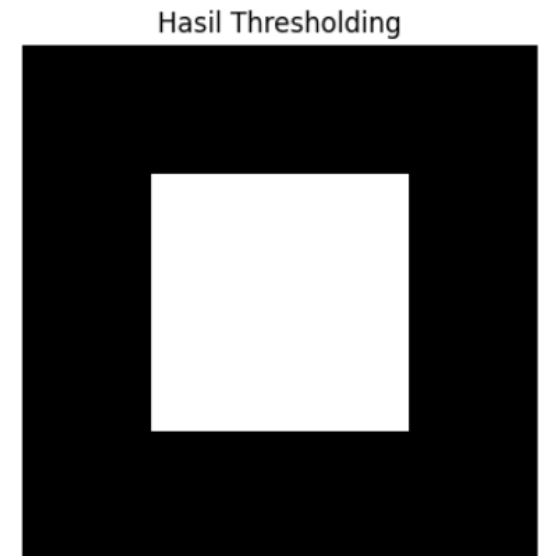
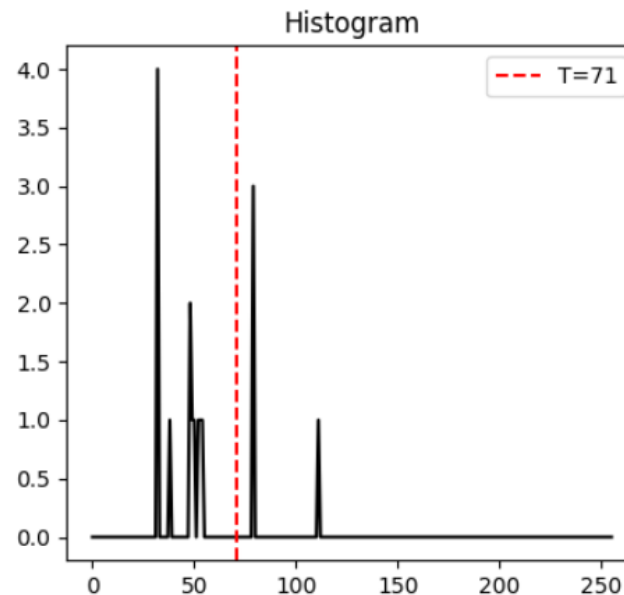
???

$$\begin{bmatrix} 32 & 48 & 32 & 54 \\ 32 & 111 & 79 & 32 \\ 38 & 79 & 79 & 48 \\ 50 & 53 & 52 & 49 \end{bmatrix}$$

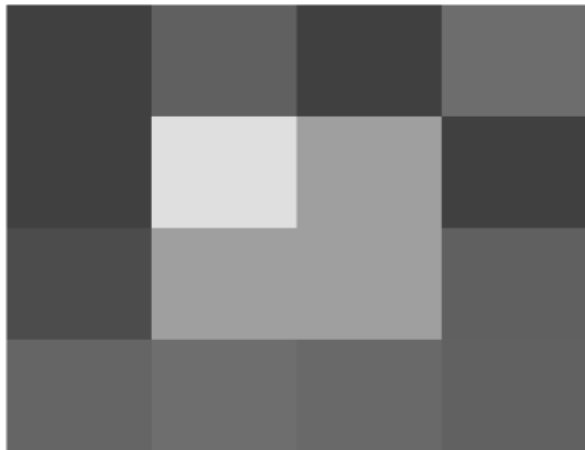

Nilai	Freq
0	0
32	4
38	1
48	2
50	1
52	1
53	1
54	1
79	3
111	1
112	0

Threshold,

$$(32 + 111) / 2 \approx 71$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 255 & 255 & 0 \\ 0 & 255 & 255 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$


Thresholding – Dua Puncak

$$\begin{bmatrix} 32 & 48 & 32 & 54 \\ 32 & 111 & 79 & 32 \\ 38 & 79 & 79 & 48 \\ 50 & 53 & 52 & 49 \end{bmatrix}$$


Nilai	Freq
32	4
38	1
48	2
50	1
52	1
53	1
54	1
79	3
111	1

- Puncak 1 $\rightarrow$ 32
- Puncak 2 $\rightarrow$ 79

Logik???
(Nilai tambahan)

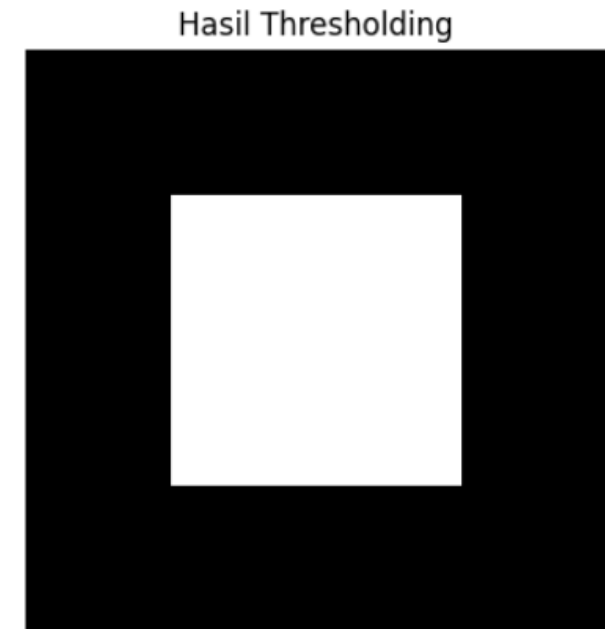
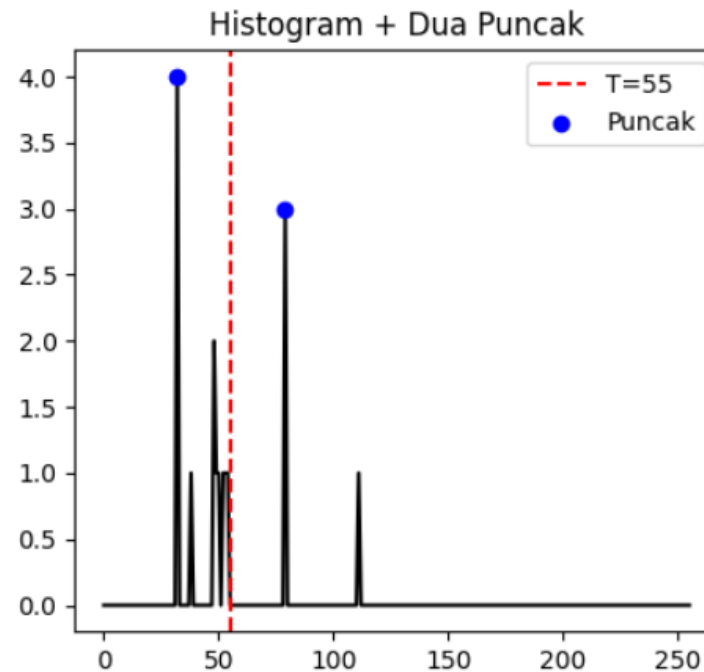
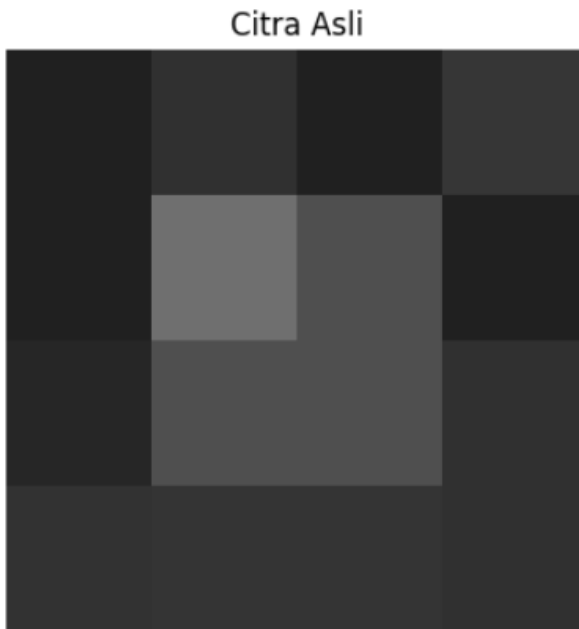
Threshold $\rightarrow (32 + 79) / 2 \approx 55$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 255 & 255 & 0 \\ 0 & 255 & 255 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thresholding - Nilai Tengah Dua Puncak

Nilai	Freq
32	4
38	1
48	2
50	1
52	1
53	1
54	1
79	3
111	1

Puncak histogram: P1=32, P2=79, Threshold=55



Bisa gak kalau ambil threshold di Lembah?

Thresholding - Lembah (bin diatas 0)

- Puncak 1 → 32
- Puncak 2 → 111

Threshold → $(38 + 111) / 2 \approx 74$

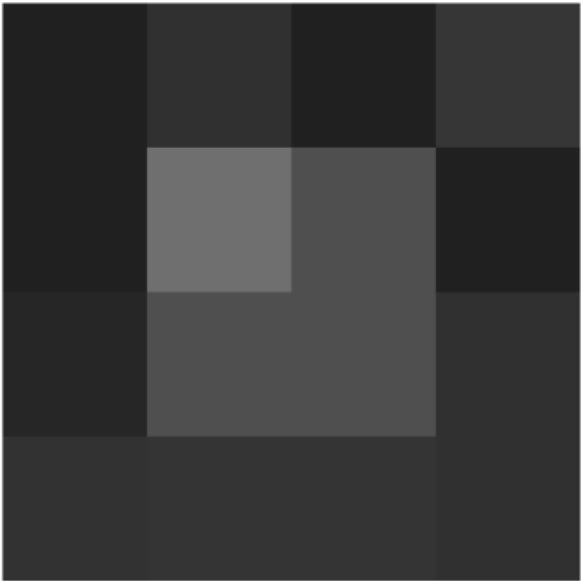
Logik yang sama dengan dua Puncak

Threshold → $(50 + 54) / 2 \approx 52$???

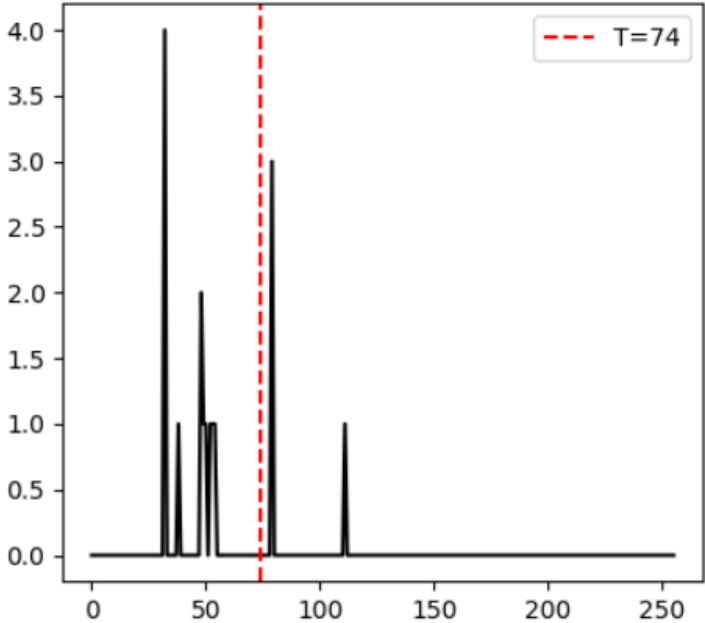
Nilai	Freq
32	4
38	1
48	2
50	1
52	1
53	1
54	1
79	3
111	1

min=32, max=111, Threshold=74

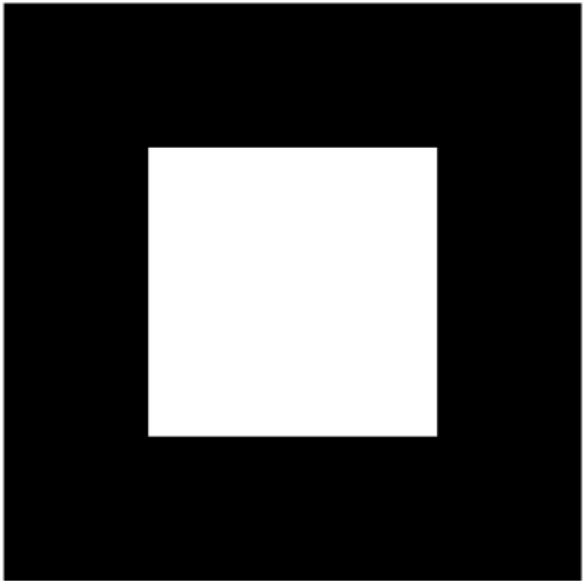
Citra Asli



Histogram



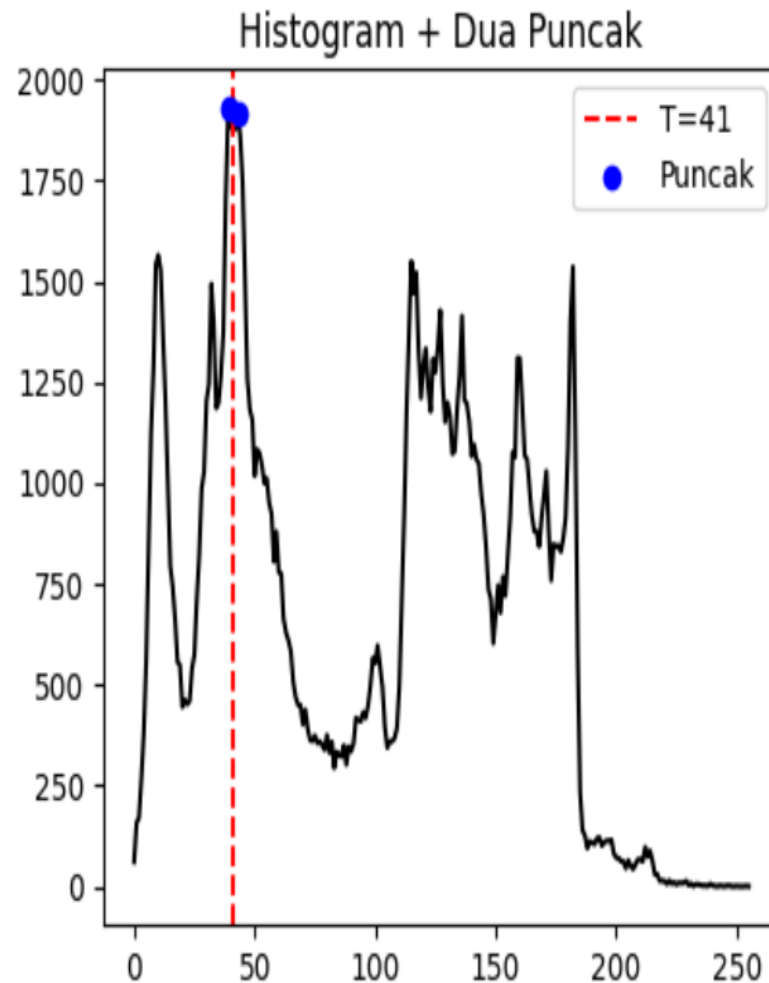
Hasil Thresholding



Thresholding - Dua Puncak

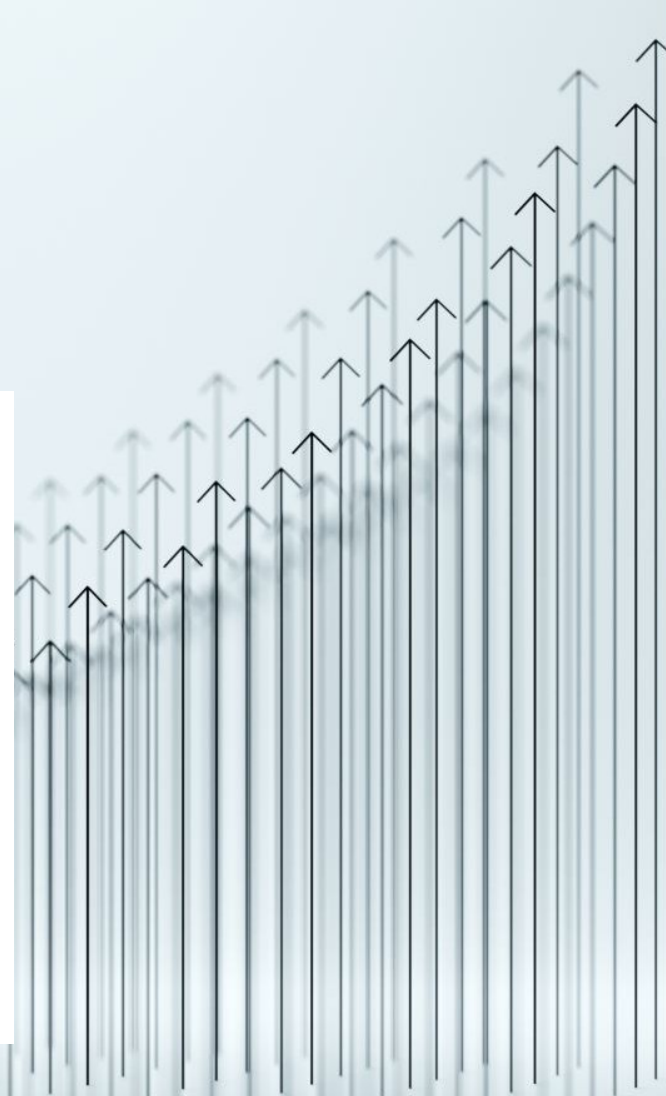
Real Image

Puncak histogram: 39, 43, Threshold = 41



Histogram Equalization

Uniform
Gaussian

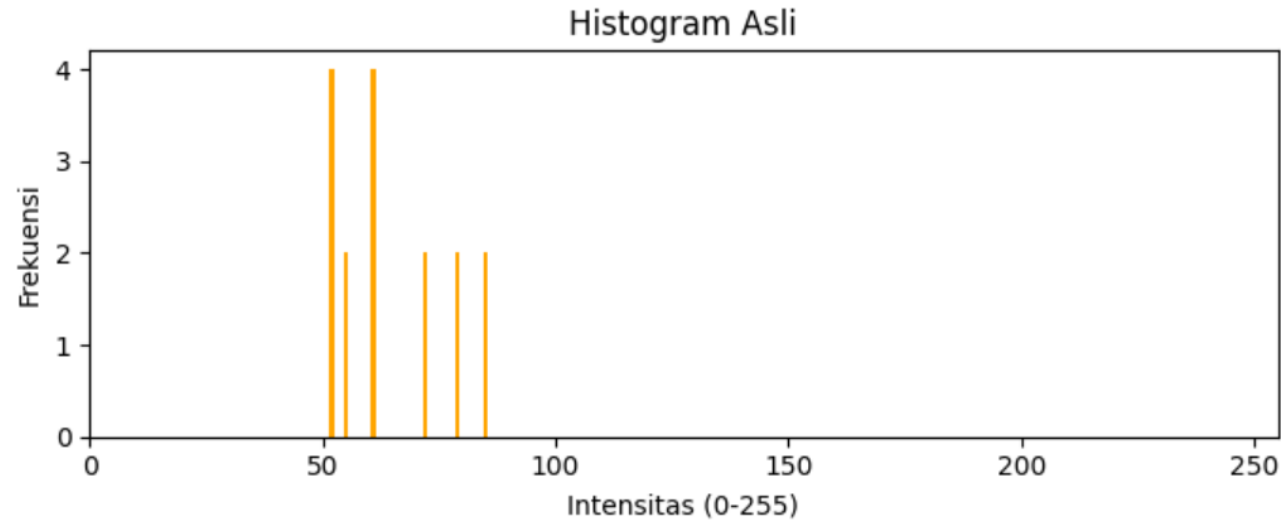


Histogram – Statistik

Citra Greyscale 4x4

52	55	52	61
52	79	61	52
85	72	61	55
61	79	72	85

Jumlah total piksel $N = 16$.



$$\bar{x} = \frac{\sum x_i}{N}$$

$$\bar{x} / \mu \text{ (mean)} = 1034/16 = 64.625$$

Intensitas
Cenderung
Rendah

$$\text{Var} = \frac{1}{N} \sum (x_i - \bar{x})^2$$

$$\text{Var} / \sigma^2 \text{ (varian)} = 2227.750/16 = 139.234$$

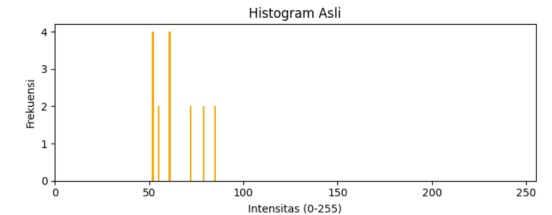
$$\sigma = \sqrt{\text{Var}}$$

$$\sigma / \text{Standard Deviasi} = \sqrt{139.234} = 11.800$$

Kontras
Rendah

Histogram Equalization – Perbaikan Kontras

$$p(r_k) = \frac{n_k}{N}$$



Citra Greyscale 4x4

52	55	52	61
52	79	61	52
85	72	61	55
61	79	72	85

Jumlah total piksel $N = 16$.

Nilai Intensitas	Frekuensi	PDF
52	4	$4/16 = 0.2500$
55	2	$2/16 = 0.1250$
61	4	$4/16 = 0.2500$
72	2	$2/16 = 0.1250$
79	2	$2/16 = 0.1250$
85	2	$2/16 = 0.1250$

PDF → **Probability Density Function** → histogram yang sudah **dinormalisasi jadi probabilitas**.

Histogram Equalization – Uniform

$$c(r_k) = \sum_{j=0}^k p(r_j)$$

Nilai Intensitas	Frekuensi	PDF	CDF
52	4	4/16 = 0.2500	$c(52) = p(52) + 0 = 0.250$
55	2	2/16 = 0.1250	$c(55) = p(52) + p(55) = 0.250 + 0.1250 = 0.375$
61	4	4/16 = 0.2500	$c(61) = 0.375 + 0.2500 = 0.625$
72	2	2/16 = 0.1250	$c(72) = 0.625 + 0.1250 = 0.750$
79	2	2/16 = 0.1250	$c(79) = 0.750 + 0.1250 = 0.875$
85	2	2/16 = 0.1250	$c(85) = 0.875 + 0.1250 = 1.000$

CDF → Cumulative Distribution Function dipakai untuk mapping di Histogram Equalization.

Mapping Intensitas – Uniform

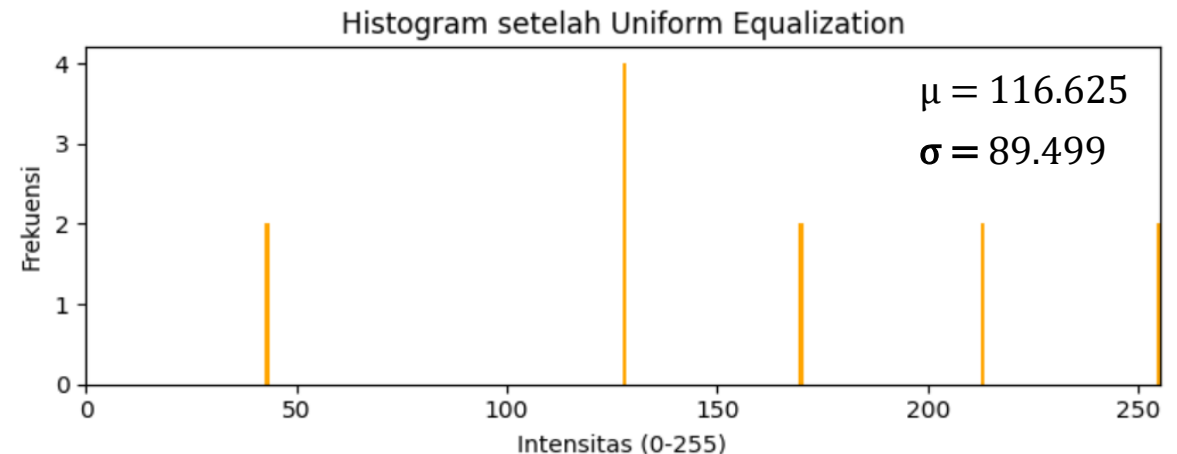
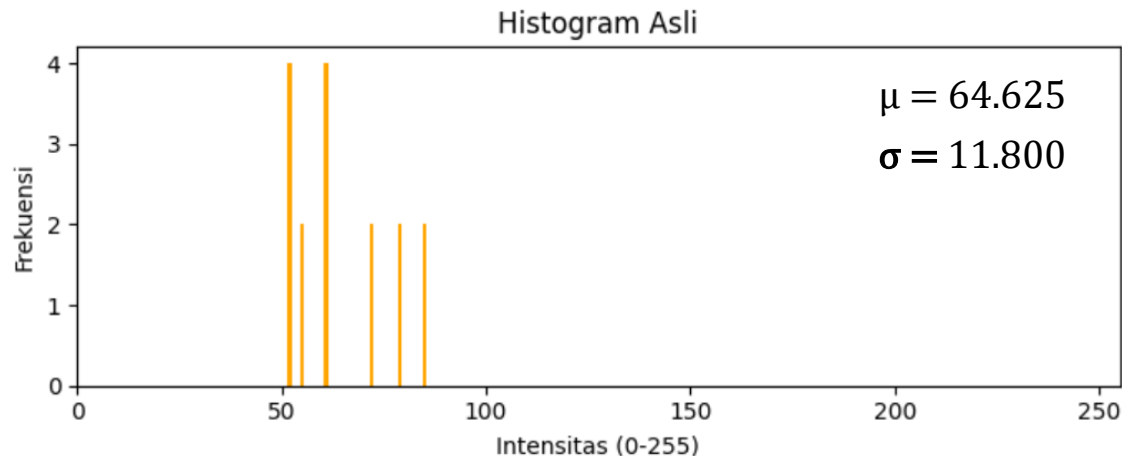
Nilai	CDF
52	0.250
55	0.375
61	0.625
72	0.750
79	0.875
85	1.000

$$s = \text{round} \left(\frac{CDF(r) - CDF_{\min}}{1 - CDF_{\min}} \cdot (L - 1) \right)$$

$\begin{bmatrix} 52 & 55 & 52 & 61 \\ 52 & 79 & 61 & 52 \\ 85 & 72 & 61 & 55 \\ 61 & 79 & 72 & 85 \end{bmatrix}$

- $52 \rightarrow (0.250 - 0.250) / (1 - 0.250) \times 255 \approx 0$
- $55 \rightarrow (0.375 - 0.250) / (1 - 0.250) \times 255 \approx 42$
- $61 \rightarrow (0.625 - 0.250) / (1 - 0.250) \times 255 \approx 127$
- $72 \rightarrow (0.750 - 0.250) / (1 - 0.250) \times 255 \approx 170$
- $79 \rightarrow (0.875 - 0.250) / (1 - 0.250) \times 255 \approx 212$
- $85 \rightarrow (1.000 - 0.250) / (1 - 0.250) \times 255 \approx 255$

$\begin{bmatrix} 0 & 42 & 0 & 127 \\ 0 & 212 & 127 & 0 \\ 255 & 170 & 127 & 42 \\ 127 & 212 & 170 & 255 \end{bmatrix}$



Mapping Intensitas – Uniform

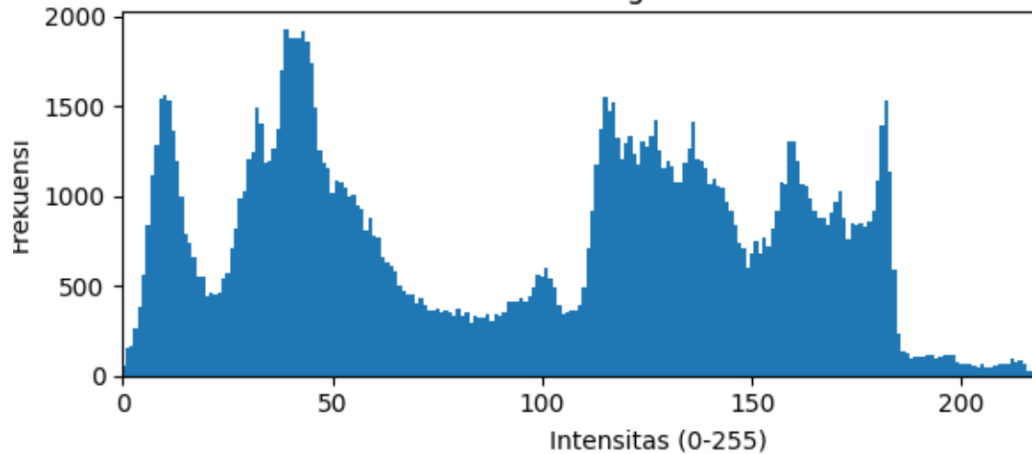
Asli (Grayscale)



Histogram Equalization (Uniform)



Histogram Asli



```
import numpy as np

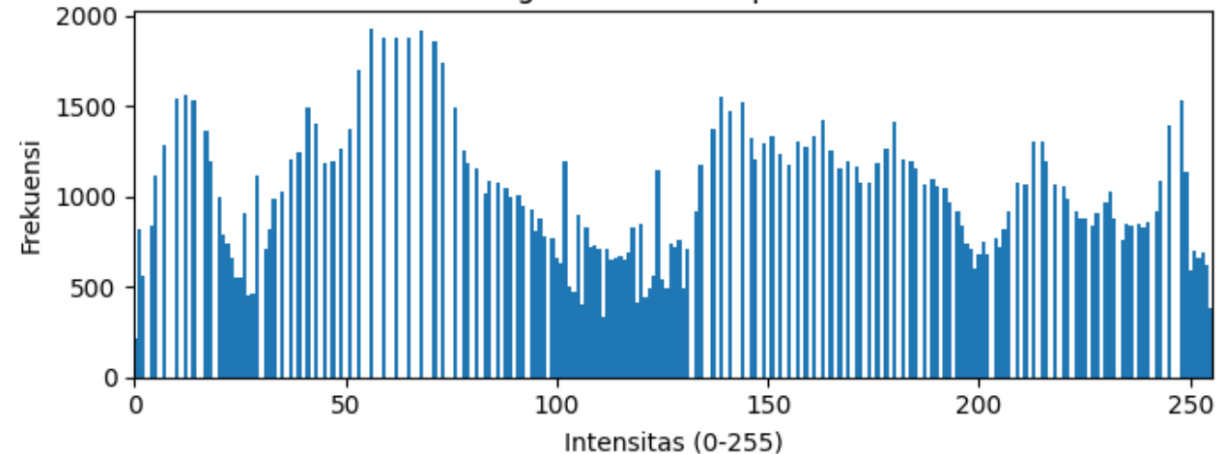
def hist_equalize(img, L=256):
    # Hitung histogram (frekuensi intensitas)
    hist = np.bincount(img.ravel(),
                        minlength=L).astype(np.float64)

    # Hitung CDF (Cumulative Distribution Function)
    cdf = np.cumsum(hist) / img.size
    cdf_min = cdf[np.nonzero(cdf)][0]

    # Hitung mapping hasil equalization
    mapping = np.floor(
        (cdf - cdf_min) / (1 - cdf_min) * (L - 1)
    )
    mapping = np.clip(mapping, 0,
                      L - 1).astype(np.uint8)

    # Terapkan mapping ke citra
    return mapping[img]
```

Histogram setelah Equalization



Mapping Intensitas – Gaussian

```
from scipy.special import erfinv
erfinv(0.250)
```

Nilai	CDF
52	0.250
55	0.375
61	0.625
72	0.750
79	0.875
85	1.000

$$p = \text{CDF}_{\text{src}}(i)$$

$$g' = \frac{g - g_{\min}}{g_{\max} - g_{\min}} \times 255$$

$$g = \mu + \sigma z, \quad z = \sqrt{2} \operatorname{erf}^{-1}(2p - 1)$$

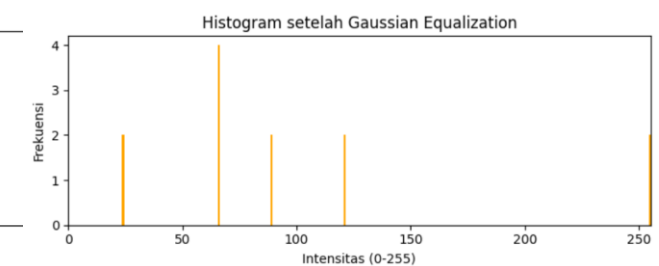
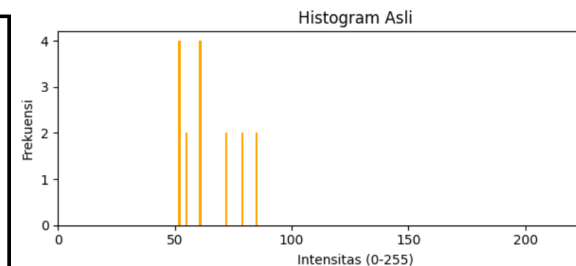
$$\mu = 128, \sigma = 40$$

Nilai	CDF	z	g	g'
52	0.250	=NORM.S.INV(0.250) = -0.674	=128 + (40 * -0.674) = 101	0
55	0.375	=NORM.S.INV(0.375) = -0.319	=128 + (40 * -0.319) = 115	24
61	0.625	=NORM.S.INV(0.625) = 0.319	=128 + (40 * 0.319) = 141	66
72	0.750	=NORM.S.INV(0.750) = 0.674	=128 + (40 * 0.674) = 155	89
79	0.875	=NORM.S.INV(0.875) = 1.150	=128 + (40 * 1.150) = 174	121
85	1.000	→	255	255

$\mu = 64.625$
 $\sigma = 11.800$

$\mu = 77.625$
 $\sigma = 77.998$

52	55	52	61	→	0	24	0	66
52	79	61	52		0	121	66	0
85	72	61	55		255	89	66	24
61	79	72	85		66	121	89	255



Mapping Intensitas – Gaussian

Asli (Grayscale)



Histogram Gaussian ($\mu=128$, $\sigma=40$)



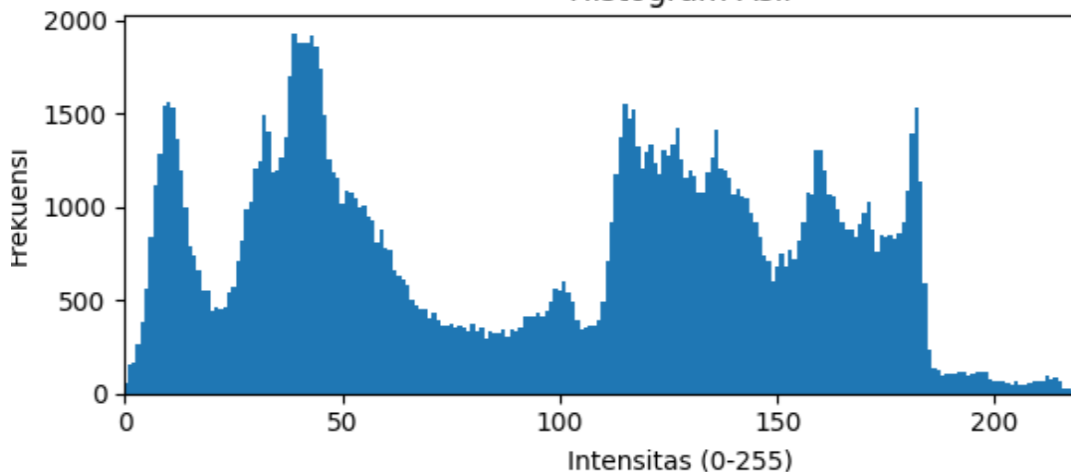
```
import numpy as np
from scipy.stats import norm
def gaussian_match(img, mu=128.0, sigma=40.0, L=256):
    # Histogram sumber
    hist = np.bincount(img.ravel(),
                        minlength=L).astype(np.float64)
    cdf_src = np.cumsum(hist) / img.size

    # CDF target (distribusi Gaussian)
    xs = np.arange(L)
    cdf_tgt = norm.cdf(xs, loc=mu, scale=sigma)
    cdf_tgt = (cdf_tgt - cdf_tgt.min())
    cdf_tgt /= cdf_tgt.max()

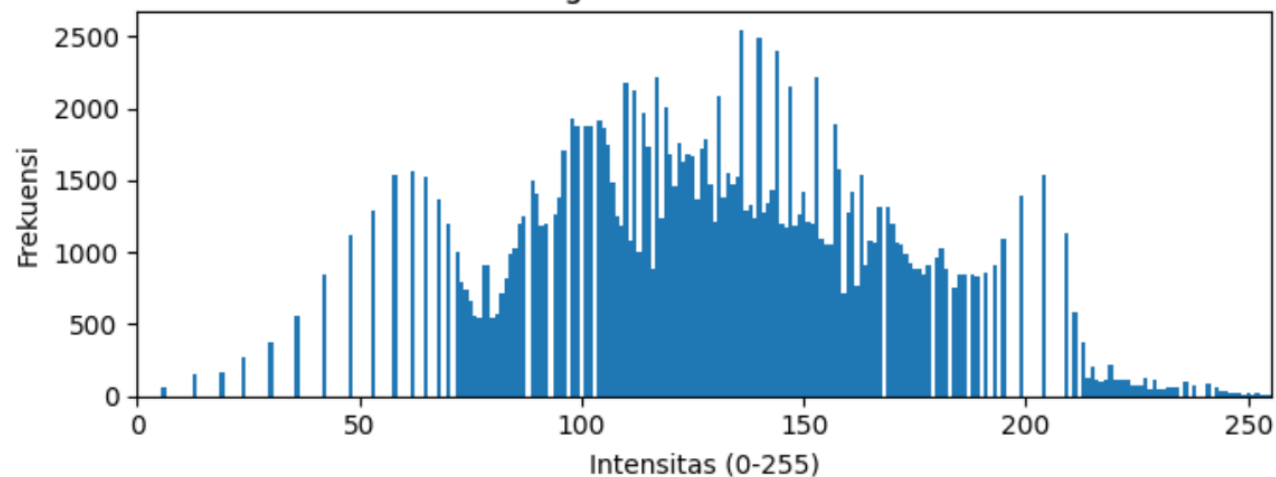
    # Mapping dari CDF sumber ke target
    mapping = np.searchsorted(cdf_tgt,
                             cdf_src, side="left")
    mapping = np.clip(mapping, 0,
                      L - 1).astype(np.uint8)

    # Terapkan mapping ke citra
    return mapping[img]
```

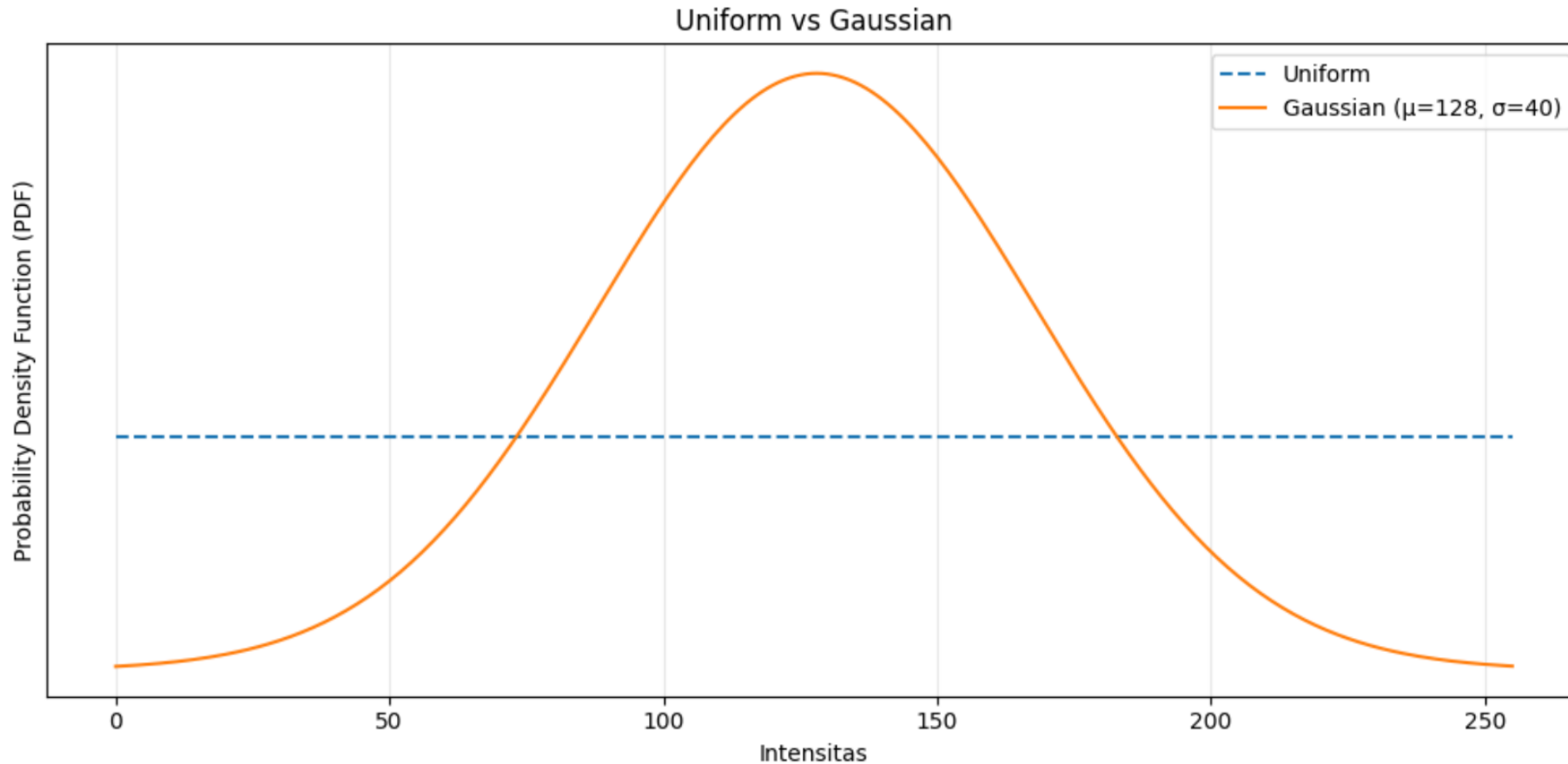
Histogram Asli



Histogram setelah Gaussian



Mapping – Uniform vs Gaussian



Gaussian

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

μ Mean	Posisi tengah, posisi puncak kurva
σ Standard Deviation	Mengontrol lebar kurva.

Mapping di RGB?

Perbaikan Kontras (Image Enhancement)

Transform RGB
ke YCbCr

Ubah nilai Y
menjadi Y_{baru}

Transform balik
 $Y_{\text{baru}} + Cb + Cr$ ke
RGB

Transform RGB
ke HSV/HSL

Ubah nilai V/L
menjadi V/L_{baru}

Transform balik
 $H + S + V/L_{\text{baru}}$ ke
RGB

YCbCr/HSV/HSL ???, Perkuliahan selanjutnya di materi Ruang Warna



Mapping	Bentuk	Efek	Kegunaan	
Uniform	Datar	Kontras global tinggi	Perbaikan Kontras	Konstan datar
Gaussian	Lonceng	Natural, lembut	Wajah, Lanskap	$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$
Exponential	Turun cepat	Tekankan gelap	Medis, Night	$p(z) = \lambda e^{-\lambda z}, \quad F(z) = 1 - e^{-\lambda z}$
Rayleigh	Setengah lonceng	Cerah natural	Outdoor, Fog	$p(z) = \frac{z}{\sigma^2} e^{-\frac{z^2}{2\sigma^2}}, \quad z > 0$
Logarithmic	Log	Detail area gelap	Night Vision	$F(z) = \frac{\log(1 + \alpha z)}{\log(1 + \alpha z_{max})}$
Sigmoid	S-Curve	Kontras tengah lembut	Tone Mapping	$F(z) = \frac{1}{1 + e^{-a(z-b)}}$
Piecewise Linear	Bebas	Sesuai kebutuhan	Sensor/Satelit	$F(z) = \begin{cases} \alpha_1 z, & z < z_1 \\ \alpha_2(z - z_1) + F(z_1), & z_1 \leq z < z_2 \\ \alpha_3(z - z_2) + F(z_2), & z \geq z_2 \end{cases}$
Laplacian	Tajam di μ	Edge emphasize	Preprocessing	$p(z) = \frac{1}{2b} e^{-\frac{ z-\mu }{b}}$
Miripkan ke citra lain	Empiris	Samakan tone	Style Transfer	Di Next Slide

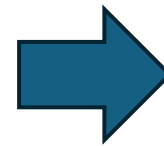
Mapping – Style Transfer

Citra Input

Nilai	CDF yang ingin diubah seperti CDF DB
0	0
50	0.15
75	0.30
100	0.35
125	0.48
150	0.6
175	0.85
200	1

Citra DB

Nilai	CDF DB
0	0
30	0.14
80	0.33
140	0.58
220	0.83
255	1



Citra Hasil Mapping

Nilai	Nilai Mapping
0	0
50	30
75	80
100	80
125	140
150	140
175	220
200	1

Cari jarak yang paling dekat → Euclidean

Tugas

- Input satu Citra kemudian :
 - Tampilkan Histogram RGB dan Greyscale dalam bentuk grafik dan angkanya.
 - Tampilkan Nilai Threshold dari dua puncak histogram kemudian ubah menjadi citra Biner berdasarkan Threshold, tampilkan citra Biner-nya.
 - Lakukan Histogram Equalization (uniform) pada citra tersebut dan Tampilkan citra hasil setelah proses Equalization (uniform). Tampilkan nilai Mean dan Standard Deviasi sebelum dan sesudah Equalization.